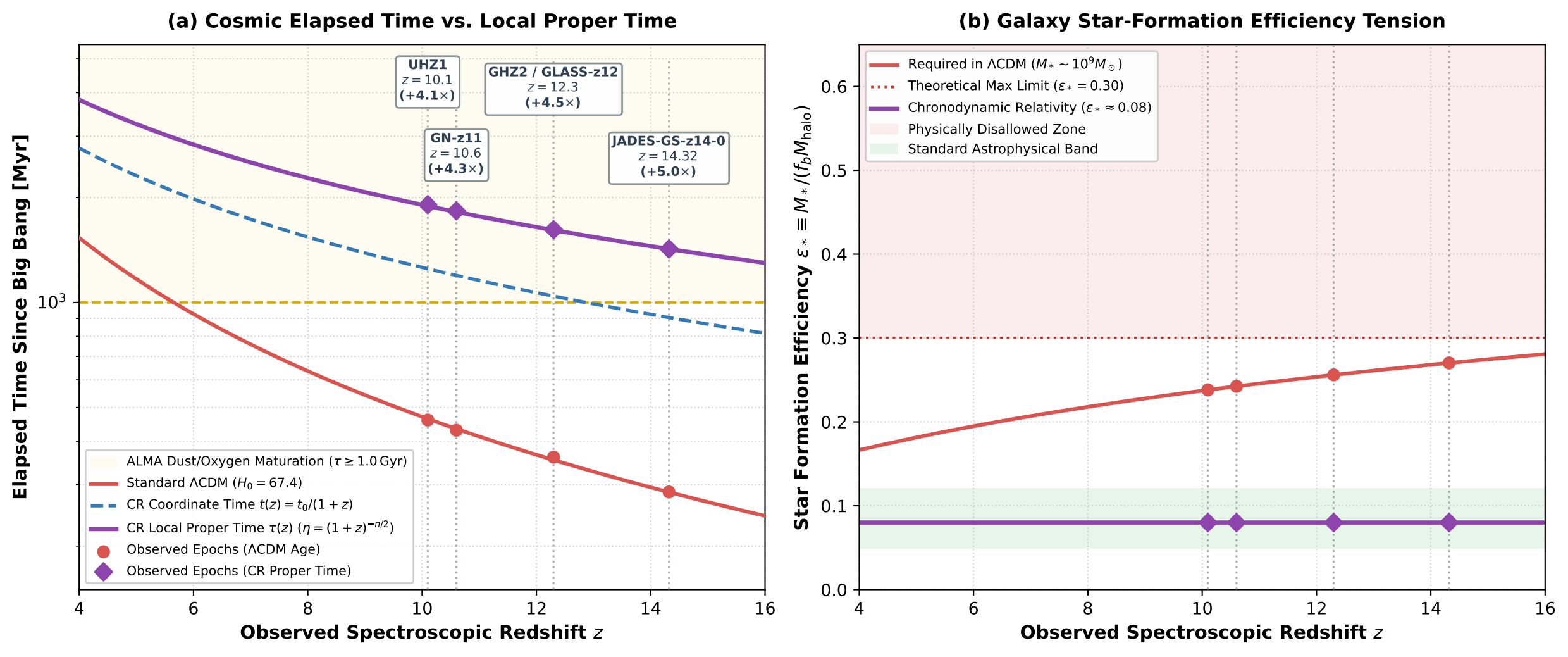
(a) Cosmic Elapsed Time vs. Local Proper Time
UHZ1
$z = 10.1$
(+4.1×)
GHZ2 / GLASS-z12
$z = 12.3$
(+4.5×)
GN-z11
$z = 10.6$
(+4.3×)
JADES-GS-z14-0
$z = 14.32$
(+5.0×)
10^3
ALMA Dust/Oxygen Maturation ($\tau \geq 1.0$ Gyr)
Standard ΛCDM ($H_0 = 67.4$)
CR Coordinate Time $t(z) = t_0/(1+z)$
CR Local Proper Time $\tau(z)$ ($\eta = (1+z)^{-n/2}$)
Observed Epochs (ΛCDM Age)
Observed Epochs (CR Proper Time)
Elapsed Time Since Big Bang [Myr]
Observed Spectroscopic Redshift z
4
6
8
10
12
14
16
(b) Galaxy Star-Formation Efficiency Tension
Required in ΛCDM ($M_* \sim 10^9 M_\odot$)
Theoretical Max Limit ($\varepsilon_* = 0.30$)
Chronodynamic Relativity ($\varepsilon_* \approx 0.08$)
Physically Disallowed Zone
Standard Astrophysical Band
Star Formation Efficiency $\varepsilon_* \equiv M_*/(f_b M_{\rm halo})$
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Observed Spectroscopic Redshift z
4
6
8
10
12
14
16